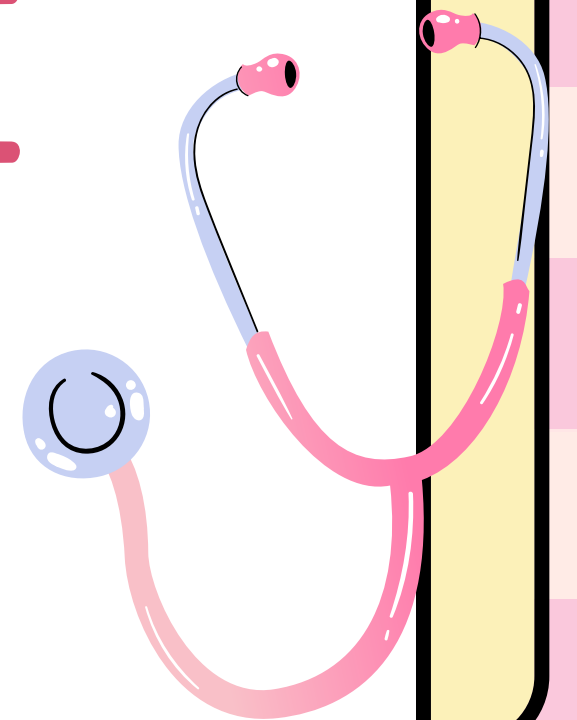


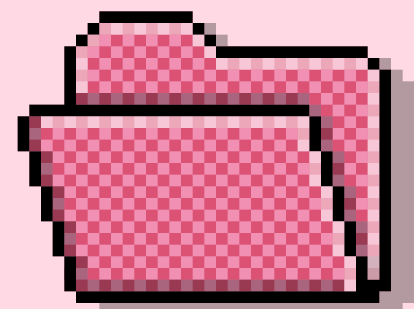


MEDICINE, MACHINE LEARNING AND MORE



ATHEER ABDULQADER ALBOLOSHI





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QUESTIONS:

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- Do you think AI can replace physicians?
- What do we need to make a good medical AI model?



ELEMENTS NEEDED TO MAKE A GOOD MEDICAL AI MODEL:

ACCURATE: AS HIGH
AS PHYSICIANS

LOW COST: LOWER
THAN A PHYSICIAN

GENERALIZABLE: ABLE
TO DO EVERYTHING A
SPECIALIST DOES,
NOT JUST ONE
NARROW TASK

CONTEXT AWARE: ABLE TO
WEIGH PATIENT-SPECIFIC
FACTORS (AGE, HISTORY,
APPEARANCE, BEHAVIOR),
NOT JUST PATTERN-MATCH
THE PRESENTING
COMPLAINT
MULTIMODAL(ALL SENSES)

TAILORED: ADAPTED
TO DIFFERENT
POPULATIONS,
GENETICS, DISEASE
PREVALENCE

EXPLAINABLE – NOT A
BLACK BOX

SELF-AWARE OF
UNCERTAINTY:
KNOWS WHAT IT
DOESN'T KNOW



EVEN THE "EASY" SPECIALTIES AREN'T SOLVED



- Pathology & radiology: the specialties AI is expected to replace first. Current pathology models beating human accuracy exist only for a subset of cancer types – not autoimmune disease, not the full range of what a pathologist actually sees daily





CRITERION: ACCURACY – THERE'S NO STABLE
"TRUTH" TO BE ACCURATE AGAINST



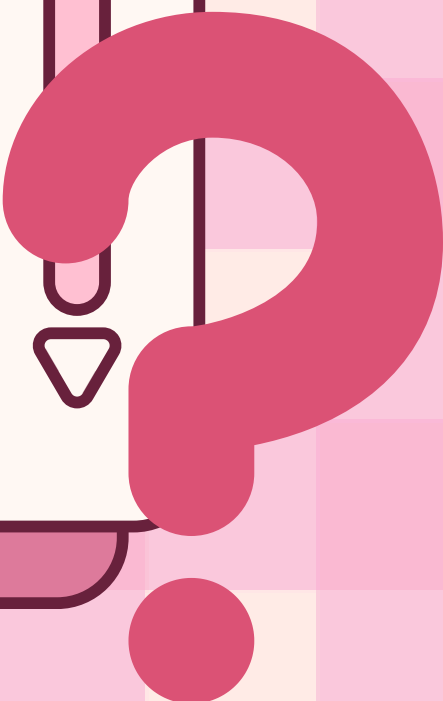
PATHOLOGIST AGREEMENT: ~50-90%
DEPENDING ON DISEASE. RADIOLOGIST
AGREEMENT: ~58-96% DEPENDING ON TASK.



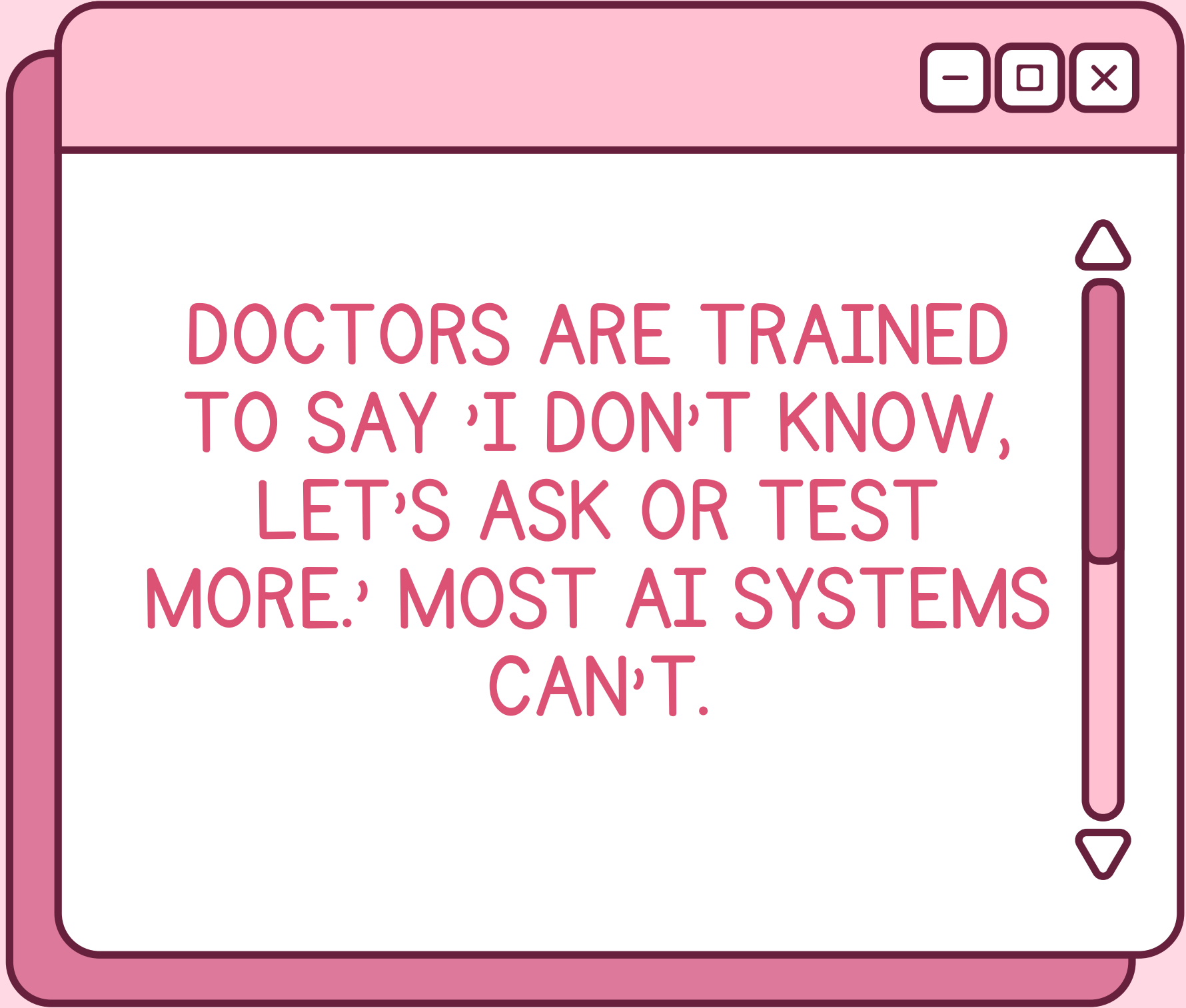
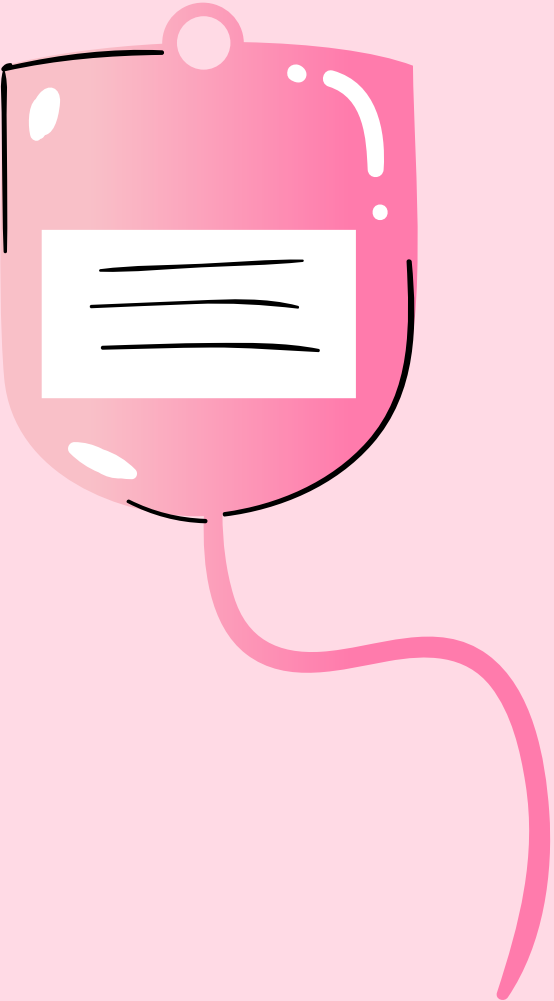
CONTEXT-
AWARE: SAME
SYMPTOM,
COMPLETELY
DIFFERENT
DIAGNOSIS



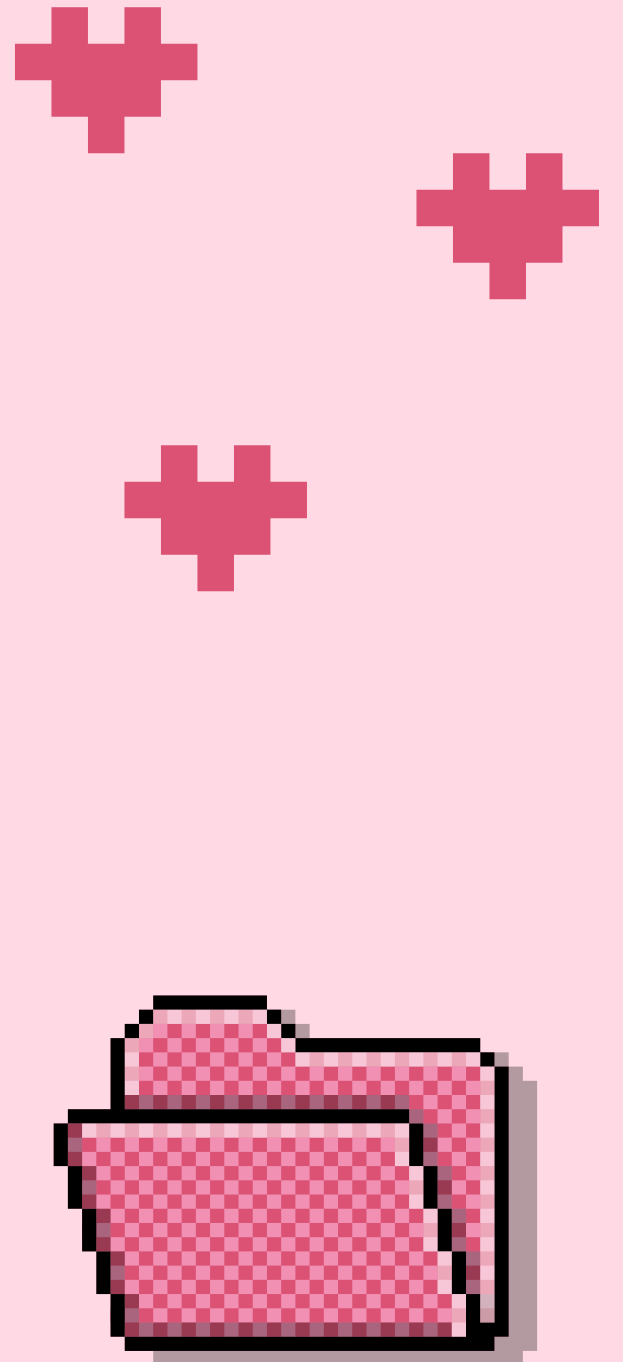
PERSISTENT DIARRHEA IN A
CHILD → INFECTION,
TODDLER'S DIARRHEA, CELIAC
DISEASE, CHLORIDE-LOSING
ENTEROPATHY,
IMMUNODEFICIENCY, OR
(RARELY) CANCER.
MULTIMODALITY



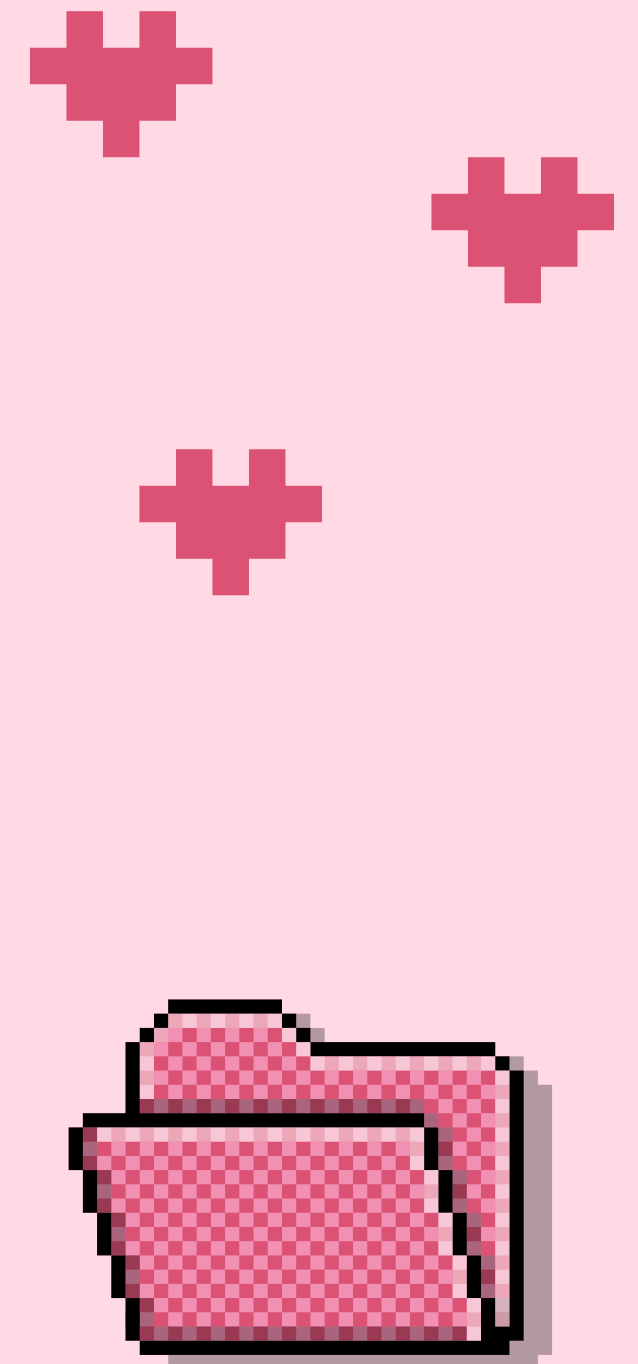
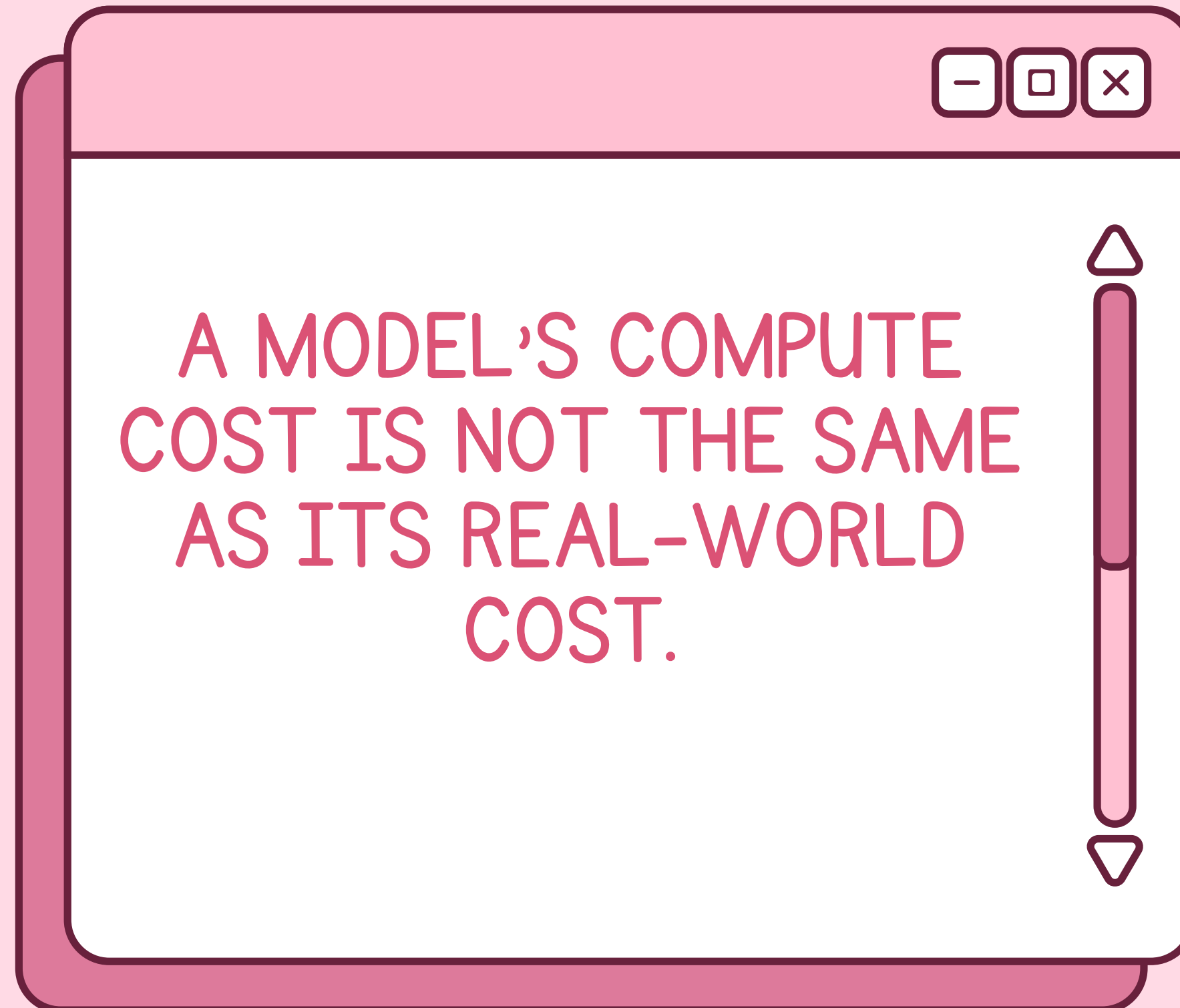
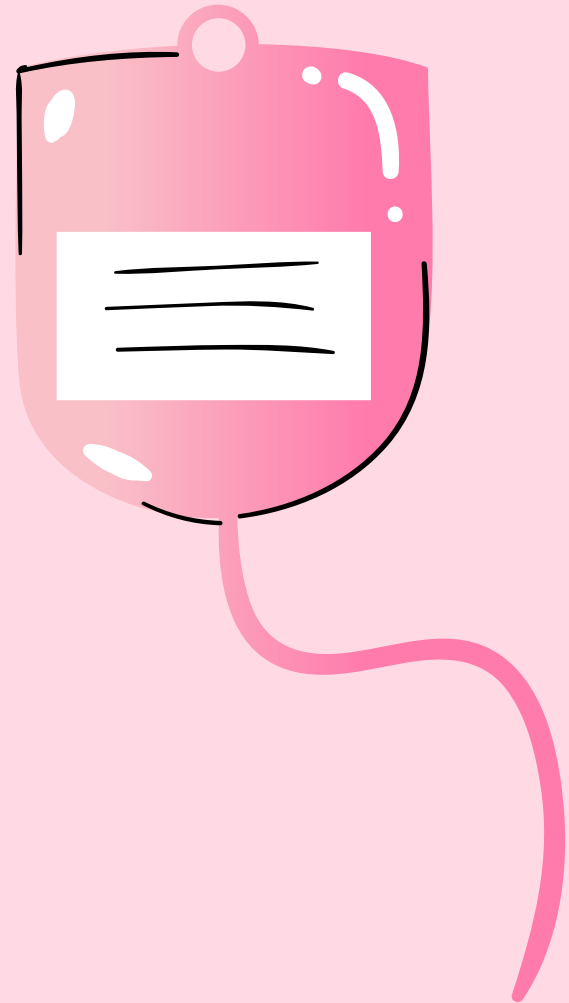
UNCERTAINTY: AI DOESN'T KNOW WHAT IT DOESN'T KNOW

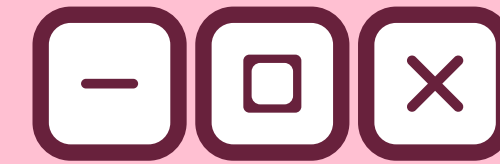


DOCTORS ARE TRAINED
TO SAY 'I DON'T KNOW,
LET'S ASK OR TEST
MORE.' MOST AI SYSTEMS
CAN'T.



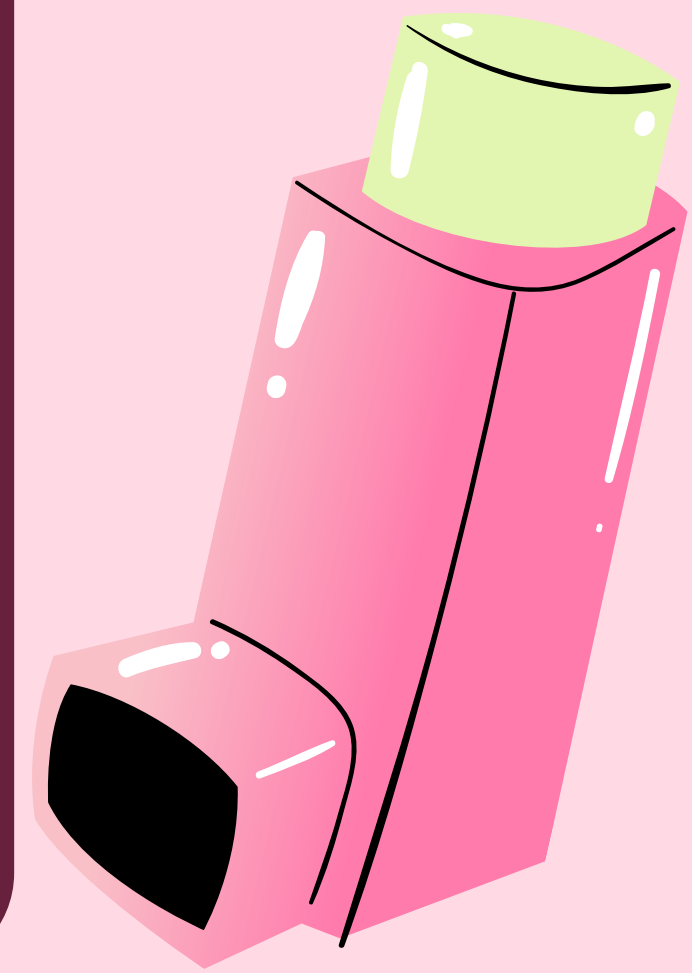
COST: CHEAP TO RUN ISN'T CHEAP TO DEPLOY

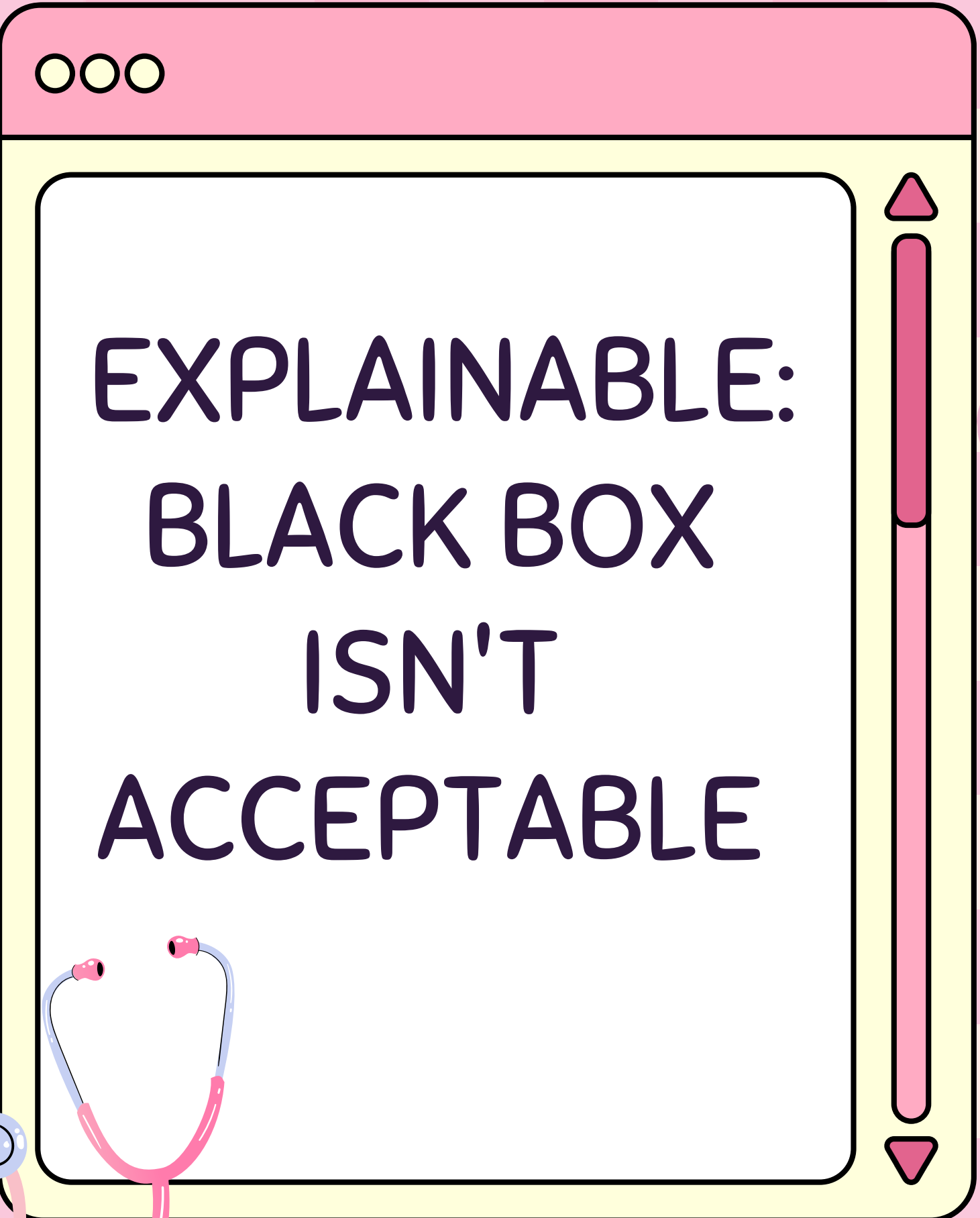




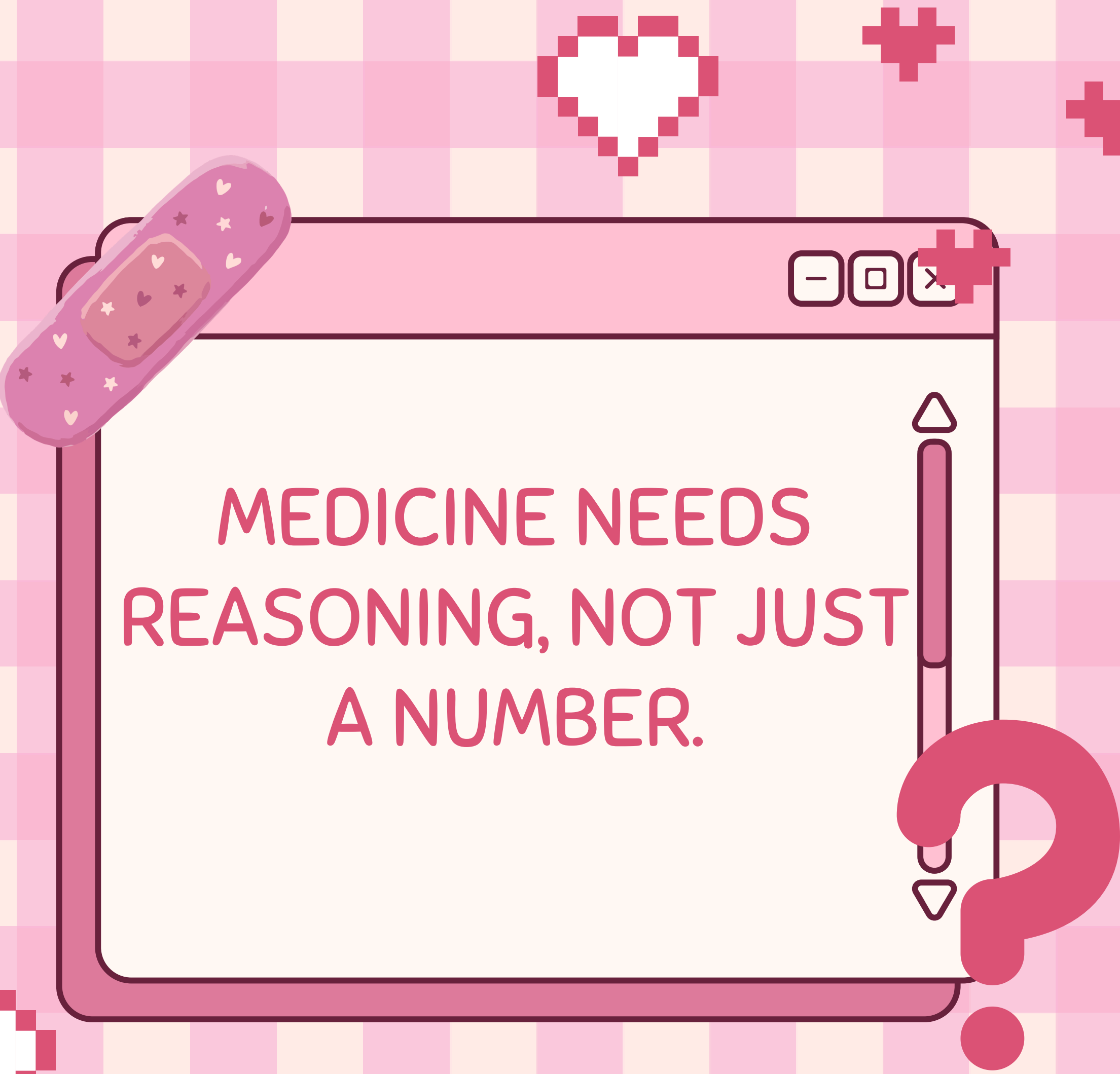
TAILORED: "BETTER DATA" ISN'T UNIVERSAL

DATA FROM SAUDI ARABIA $\neq$ DATA FROM
NORTH AMERICA $\neq$ DATA FROM SOUTH
AMERICA – DIFFERENT GENETICS, DIFFERENT
DISEASE PREVALENCE.

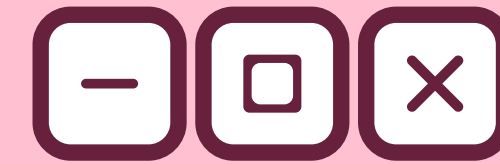




EXPLAINABLE:
BLACK BOX
ISN'T
ACCEPTABLE



MEDICINE NEEDS
REASONING, NOT JUST
A NUMBER.



THE HUMANE ELEMENT

PATIENTS DON'T JUST NEED A CORRECT ANSWER. THEY NEED TO FEEL SEEN." STUDIES SHOW MENTIONING AI USE LOWERS PATIENTS' TRUST AND PERCEIVED EMPATHY OF THEIR DOCTOR – EVEN WHEN THE MEDICAL CONTENT IS IDENTICAL.

